# Notes: Kinnan & Townsend (AER P&P, 2012)
## Kinship and Financial Networks, Formal Financial Access, and Risk Reduction

## Contents

# 1 Big picture

- **Question.** How do kinship networks and financial networks help rural households reduce risk? More specifically, which network matters for consumption smoothing, and which network matters for investment financing?
- **Core idea.** Consumption smoothing and investment smoothing require different types of insurance. Small, frequent deviations between income and desired consumption may be handled by bank-linked financial networks. Large investment opportunities may require strong social collateral, such as kinship ties.
- **Setting.** The paper studies 531 households in 16 Thai villages observed monthly from 1999 to 2005 in the Townsend Thai Monthly Survey.
- **Financial network measure.** The authors use within-village gifts, borrowing, and lending to construct directed financial links. A household can be directly connected to a bank or indirectly connected through another household that is connected to a bank.
- **Kinship measure.** A household has local kin if parents, siblings, adult children, or parents' siblings of the household head or spouse live in the same village.
- **Consumption result.** Direct and indirect access to banks are almost equally effective at smoothing consumption against income shocks. Households with no bank connection are much less insured.
- **Investment result.** Kinship ties, not bank links, are the key force in reducing investment-cash-flow sensitivity. The effect is concentrated in occupations with large investment needs relative to net worth.
- **Spillover implication.** Measuring formal financial access only by direct bank use is misleading. Households may benefit from formal finance through village lending chains.
- **Economic takeaway.** Formal financial networks work well for collateralizable, relatively small consumption-smoothing needs. Kin networks work better for large, hard-to-collateralize investment financing needs, where social sanctions and repeated relationships support repayment.

# 2 Data and measurement

## 2.1 Household data

- Data come from the monthly Townsend Thai Survey, 1999-2005.
- The sample contains 531 households in 16 villages observed for 84 months.
- The data record demographics, income, expenditure, occupation, assets, transfers, borrowing, and lending.
- The most common occupation in the sample is rice farming, followed by nonagricultural labor or business, corn growing, livestock, and agricultural wage labor.

**Interpretation:** The data are unusually rich because they observe both high-frequency economic outcomes and detailed village-level financial links.

## 2.2 Financial network data

- The authors use gifts, loans, and transfers with other households in the same village.
- For each transaction, the surveyed household identifies the counterparty household's structure or address.
- Counterparties are matched to a village census, so the authors can identify within-village financial links even when the counterparty is not itself in the monthly survey.
- Links are directed from lender/giver to borrower/receiver because the object of interest is access to funds.

- Time variation in links is collapsed: if household $i$ ever borrows from or receives a transfer from household $j$, that link is treated as present.

**Interpretation:** A household may benefit from expected access to funds even in months when it does not borrow. Collapsing the network captures the insurance value of potential access, not just realized transactions.

## 2.3 Network distance and bank access

For households $i$ and $j$, let

$$\gamma_{ij} = \text{length of the shortest directed path from } i \text{ to } j.$$

Household $i$ is reachable by household $j$ when

$$r_{ij} = 1\{\gamma_{ij} < \infty\}.$$

For banks, define

$$\gamma_{iB} = \text{length of the shortest directed path from household } i \text{ to a bank.}$$

Then

$$d_{iB} = 1\{\gamma_{iB} = 1\}, \qquad r_{iB} = 1\{\gamma_{iB} < \infty\}.$$

**Interpretation:** $d_{iB}$ captures direct borrowing from a bank. $r_{iB}$ captures any direct or indirect path to bank finance. An indirect connection means a household can access bank capital through other households.

## 2.4 Kinship network data

- The authors observe the location of parents, siblings, adult children, and parents' siblings of the household head and spouse.
- If any of these relatives live in the same village, the household is coded as having kin in the village:

$$k_i = 1.$$

- If no such relatives live in the village, then $k_i = 0$.
- About 74 percent of households have at least one such local relative.

**Interpretation:** Kin is a measure of social collateral. Relatives can enforce repayment through repeated interaction, social punishment, and loss of high-value family relationships.

## 2.5 Descriptive network facts

- Gifts from other households in the same village equal about 9 percent of average expenditure.
- For the 411 households that ever borrow or receive gifts from another village household, the average household borrows or receives from 3.2 other households.
- Conditional on ever borrowing, the average total amount borrowed from other village households over the seven-year sample is 73,727 baht.
- The average amount borrowed per transaction is 12,200 baht, about 60 percent of average monthly household expenditure.
- Households that ever borrow from villagers do so about 4.75 times over 84 months.

**Interpretation:** Intravillage financial transactions are not tiny symbolic transfers. They are large enough to matter for insurance and investment.

# 3 Empirical specifications

## 3.1 Consumption-smoothing specification

The consumption regression modifies Townsend's standard full-insurance test by allowing income shocks to matter differently depending on bank access, kinship, and net worth:

$$\Delta c_{ivt} = \alpha_1 \Delta y_{ivt} + \alpha_2 \Delta y_{ivt} \times d_{iB} + \alpha_3 \Delta y_{ivt} \times r_{iB} + \alpha_4 \Delta y_{ivt} \times k_i + \alpha_5 \Delta y_{ivt} \times \bar{w}_i + \delta_{B,t} + \varepsilon_{it}. \tag{1}$$

- $c_{ivt}$ is per-capita consumption of household $i$ in village $v$ and month $t$.
- $y_{ivt}$ is per-capita income.
- $d_{iB}$ indicates direct bank access.
- $r_{iB}$ indicates any path to the bank, direct or indirect.
- $k_i$ indicates local kin.
- $\bar{w}_i$ is average net worth.
- $\delta_{B,t}$ is a time effect for households connected to the financial system.

**Interpretation:** The key test is whether consumption responds to income shocks. Lower income sensitivity means better insurance. The interaction terms ask whether bank access, indirect network access, kin, and net worth reduce that sensitivity.

## 3.2 Investment-smoothing specification

For investment, the authors examine positive investment events and scale investment and income by household assets:

$$\left(\frac{I}{A}\right)_{ivt} = \alpha_1 \left(\frac{y}{A}\right)_{ivt} + \alpha_2 \left(\frac{y}{A}\right)_{ivt} \times r_{iB} + \alpha_3 \left(\frac{y}{A}\right)_{ivt} \times k_i + \alpha_4 \left(\frac{y}{A}\right)_{ivt} \times \bar{w}_i + \beta_1 r_{iB} + \beta_2 k_{iB} + \beta_3 \bar{w}_i + \delta_v + \delta_{B,t} + \varepsilon_{it}. \tag{2}$$

- $I$ is investment.
- $A$ is household assets.
- The outcome is positive investment scaled by assets.
- The income variable is cash flow or income scaled by assets.
- Village fixed effects $\delta_v$ absorb persistent local characteristics such as rainfall, suitability for occupations, and distance to towns.
- Heteroskedasticity-robust standard errors are used.

**Interpretation:** Investment-cash-flow sensitivity measures financing constraints. If investment moves strongly with current income, households cannot finance investment opportunities independently of cash on hand.

### 3.3 Why first differences and fixed effects matter

- Consumption regressions use first differences, which remove time-invariant household characteristics.
- The authors do not include household fixed effects in the investment regression because there are few positive investment events per household.
- Investment regressions include village fixed effects and a time effect for households connected to the financial system.

**Interpretation:** The design is not randomized, but the authors try to remove stable household characteristics and common shocks that would otherwise confound risk-sharing estimates.

# 4 Identification and empirical interpretation

### 4.1 What variation identifies the results

- The consumption estimates compare how income shocks translate into consumption changes for households with different network positions.
- The investment estimates compare how cash-flow shocks translate into investment changes for households with and without kin or bank network access.
- The key economic objects are interaction effects: income shock $\times$ direct bank access, income shock $\times$ any bank path, and income shock $\times$ kin.

**Interpretation:** The paper does not simply ask whether connected households consume or invest more. It asks whether they are less exposed to income fluctuations.

### 4.2 Why indirect bank connections matter

- Some households borrow directly from banks.
- Others borrow from households that borrow from banks.
- If indirect access is effective, then direct bank use is an incomplete measure of formal financial access.
- Ignoring indirect paths can misestimate the value of formal financial institutions.

**Interpretation:** The village financial network can distribute formal bank capital beyond the households that formally borrow.

### 4.3 Why kin and banks may solve different problems

- Consumption smoothing involves relatively small deviations between realized income and desired contemporaneous consumption.
- Investment financing can involve large gaps between a profitable opportunity and the household's cash on hand.
- Bank-linked financial networks may be effective for smaller, collateralizable needs.
- Kin networks may be effective for large, hard-to-collateralize investments because social punishment supports repayment.

**Interpretation:** The paper's central distinction is not financial versus social networks in general. It is which network is better suited to which type of risk.

### 4.4 Limitations

- Network measurement is incomplete: if two nonsurvey households connect a chain, the observed network may miss that path.
- The authors note that this measurement error can bias estimated indirect-link effects toward zero.
- Financial access and kinship are not randomly assigned.
- Consumption results are discussed in text rather than shown in a table.
- Investment size is endogenous, so the occupation split is used as a proxy for the scale of investment opportunities.

**Interpretation:** The results are best read as evidence on mechanisms and network channels, not as a randomized treatment effect of kin or bank access.

## 5 Original equation panels and table

The paper has no standalone figures. To preserve the original visual material, I directly crop the two key empirical specification panels and Table 1 from the paper.

### 5.1 Equation panels: consumption and investment specifications

$$
\begin{aligned}
(1)\quad \Delta c_{ivt} = {} & \alpha_1 \Delta y_{ivt} + \alpha_2 \Delta y_{ivt} \times d_{i,B} \\
& + \alpha_3 \Delta y_{ivt} \times r_{i,B} + \alpha_4 \Delta y_{ivt} \times k_i \\
& + \alpha_5 \Delta y_{ivt} \times \overline{w}_i + \delta_{B,t} + \varepsilon_{it},
\end{aligned}
$$

**(a)** Original consumption-smoothing specification.

$$
\begin{aligned}
(2)\quad \left(\frac{I}{A}\right)_{ivt} = {} & \alpha_1 \left(\frac{y}{A}\right)_{ivt} + \alpha_2 \left(\frac{y}{A}\right)_{ivt} \\
& \times r_{i,B} + \alpha_3 \left(\frac{y}{A}\right)_{ivt} \times k_i \\
& + \alpha_4 \left(\frac{y}{A}\right)_{ivt} \times \overline{w}_i + \beta_1 r_{i,B} \\
& + \beta_2 k_{i,B} + \beta_3 \overline{w}_i + \delta_v \\
& + \delta_{B,t} + \varepsilon_{it}.
\end{aligned}
$$

**(b)** Original investment-smoothing specification.

**Figure 1:** Original equation panels from Kinnan and Townsend.

**Read:**

- Equation (1) asks whether consumption changes with income changes, and whether the sensitivity differs by direct bank access, any bank path, kin, and net worth.
- Equation (2) asks whether investment scaled by assets responds to income scaled by assets, again allowing the slope to vary with bank access, kin, and wealth.

- Both equations focus on slope changes, not just level differences.

**Economic message:** The paper's core evidence comes from interaction terms. The sign and magnitude of these interactions reveal which network reduces exposure to income volatility.

## 5.2 Table 1: Kinship, financial access, and investment

TABLE 1—KINSHIP, FINANCIAL ACCESS, AND INVESTMENT

| | No controls (1) | All house-holds (2) | Above-median investment size (3) | Below-median investment size (4) |
|---|---|---|---|---|
| Income | 0.1078* | 0.6526*** | 0.6370*** | 0.0077 |
| | (0.0649) | (0.1950) | (0.2102) | (0.3359) |
| Income | | | | |
| X Any link to bank | | −0.1268 | −0.0821 | 0.2931 |
| | | (0.1288) | (0.1292) | (0.3983) |
| X Kin in village | | −0.4136*** | −0.5056*** | 0.4543 |
| | | (0.1549) | (0.1599) | (0.3256) |
| X Net worth (mill. baht) | | −0.1087 | −0.0405** | −0.3710 |
| | | (0.0762) | (0.0205) | (0.2357) |
| Observations | 6,055 | 5,794 | 2,319 | 3,463 |

*Note:* Heteroskedasticity-robust standard errors in parentheses.
***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

**Figure 2:** Original Table 1 from the paper.

**Read:**

- Column (1) shows the unconditional investment-cash-flow sensitivity: a 1 baht income increase raises investment by 0.1078 baht.
- Column (2) shows that, after interactions are added, households without kin and without bank access have a large investment response to income: 0.6526.
- In column (2), kin in the village reduces the investment response by 0.4136, significant at the 1 percent level.
- Bank network access does not significantly reduce investment-cash-flow sensitivity in the full sample: the interaction is -0.1268 and insignificant.
- Column (3) shows that the kin effect is strongest for occupations with above-median investment-to-net-worth ratios: kin reduces sensitivity by 0.5056, significant at the 1 percent level.
- Column (4) shows no meaningful kin effect for smaller-scale investment occupations.

**Economic message:** Kin networks are especially important when investment needs are large relative to wealth. This is consistent with kinship acting as social collateral for transactions too large or too hard to collateralize through ordinary bank-linked channels.

# 6 Results by outcome

## 6.1 Consumption smoothing

- A restricted version of the consumption equation gives an average consumption-income sensitivity of 0.0078.
- Households with no bank connection have much worse smoothing: a 1 baht income change is associated with a 0.1645 baht consumption change.
- Direct bank access reduces that sensitivity by 0.1658, leaving a net sensitivity of -0.0013, statistically indistinguishable from zero.
- Indirect bank access reduces sensitivity by 0.1643, leaving a net sensitivity of 0.0002, also statistically indistinguishable from zero.
- Net worth reduces consumption-income sensitivity, but the economic magnitude is small.
- Conditional on financial access and net worth, kin slightly increases consumption sensitivity by 0.0102.

**Interpretation:** Formal financial access, including indirect access through the village network, is the main consumption-smoothing mechanism. Local kin are not the central channel for small consumption-risk deviations.

## 6.2 Investment smoothing

- Investment is highly sensitive to cash flow for households without local kin.
- Kin in the village substantially reduces this sensitivity.
- Bank links do not significantly reduce investment-cash-flow sensitivity.
- The kin effect is strongest in occupations where investment opportunities are large relative to wealth.

**Interpretation:** Investment opportunities often require larger transfers than ordinary consumption insurance. Kinship may sustain repayment through social sanctions and relationship value when formal collateral is insufficient.

# 7 Short summary

- The paper studies 531 households in 16 Thai villages from 1999 to 2005.
- It builds village financial networks from gifts, loans, and transfers, and distinguishes direct from indirect bank access.
- It measures kinship by whether close relatives of the household head or spouse live in the same village.
- The consumption specification tests whether consumption moves with income shocks and whether this sensitivity is reduced by bank access, kinship, or net worth.
- The investment specification tests whether investment moves with cash flow and whether this sensitivity is reduced by bank access, kinship, or net worth.
- Direct and indirect bank access both smooth consumption almost completely.

- Kinship does not improve consumption smoothing once financial access and net worth are controlled for.
- Kinship reduces investment-cash-flow sensitivity, especially for occupations with large investment needs relative to wealth.
- Bank links do not significantly smooth investment in Table 1.
- The main lesson is that different networks insure different risks: formal financial networks help with consumption, kinship networks help with large investment financing.

# 8 Conclusion

## 8.1 Core conclusion

Kinnan and Townsend show that kinship networks and financial networks are not substitutes in a simple sense. They solve different problems. Formal financial access, even when indirect through village lending chains, is highly effective for consumption smoothing. Kinship is more important for smoothing investment when the investment opportunity is large relative to household wealth.

## 8.2 Economic intuition

Consumption shocks are relatively small and frequent. They can be smoothed through borrowing that is directly or indirectly connected to the formal financial system. Large investments are different. They may require loan sizes too large for tangible collateral or standard financial relationships. Kinship can support these transfers because the threat of losing family relationships or social standing helps enforce repayment.

## 8.3 Takeaway

The paper's key contribution is to separate two forms of network-based risk reduction. Financial networks transmit bank liquidity through villages and smooth consumption. Kin networks provide social collateral for larger investment transactions. Policy evaluations that look only at direct bank use may miss large spillovers through local networks.